

MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 10: Friday 11 September 2026

(Week 5)

Remark. Q09.8 to Q09.10 are leftover exercises from the Tuesday class.

Remark. Q10.1 to Q10.9 are on on Sections 13.4, 13.5, 13.6 and 13.9.



For a function $f(\mathbf{x}, \mathbf{y})$, the gradient of $f(\mathbf{x}, \mathbf{y})$ is the vector-valued function

$$\nabla f(\mathbf{x}, \mathbf{y}) = \langle f_x, f_y \rangle = f_x \mathbf{i} + f_y \mathbf{j} = \frac{\partial f}{\partial \mathbf{x}} \mathbf{i} + \frac{\partial f}{\partial \mathbf{y}} \mathbf{j} = \begin{pmatrix} f_x \\ f_y \end{pmatrix}$$

∇f is also read as 'del f '.

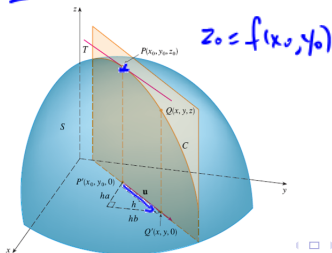
Similarly, for a function $f(\mathbf{x}, \mathbf{y}, \mathbf{z})$, its gradient is:

$$\nabla f(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \langle f_x, f_y, f_z \rangle = f_x \mathbf{i} + f_y \mathbf{j} + f_z \mathbf{k} = \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix}$$

The **directional derivative** of $f(x, y)$ at (x_o, y_o) in the direction of unit vector $\mathbf{u} = \langle \mathbf{a}, \mathbf{b} \rangle$ is $\|\mathbf{u}\| = 1$.

$$D_{\mathbf{u}}f(x_o, y_o) = \lim_{h \rightarrow 0} \frac{f(x_o + h\mathbf{a}, y_o + h\mathbf{b}) - f(x_o, y_o)}{h}. \quad (*)$$

Geometrically, let $\mathbf{P} = (x_o, y_o, f(x_o, y_o))$, and let $\mathbf{C}$ be the curve obtained as the intersection of the surface $z = f(x, y)$ with the vertical plane through $\mathbf{P}$ in the direction given by $\mathbf{u}'$. Then $D_{\mathbf{u}}f(x_o, y_o)$ is the slope of the tangent line to the curve $\mathbf{C}$ at $\mathbf{P}$.



- If $\mathbf{u} = \mathbf{i} = \langle 1, 0 \rangle$, then $D_{\mathbf{i}}f = f_x$.
- If $\mathbf{u} = \mathbf{j} = \langle 0, 1 \rangle$, then $D_{\mathbf{j}}f = f_y$.
- We have

$$D_{\mathbf{u}}f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \mathbf{u}.$$

Justification: Recall linear approximation:

$$\Delta f \approx f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y$$

Then with $\Delta \mathbf{x} = \mathbf{ha}$ and $\Delta \mathbf{y} = \mathbf{hb}$, we have

$$\begin{aligned} \frac{f(x_0 + \mathbf{ha}, y_0 + \mathbf{hb}) - f(x_0, y_0)}{h} &= \frac{\Delta f}{h} \\ &\approx \frac{f_x(x_0, y_0) \cdot \cancel{ha} + f_y(x_0, y_0) \cdot \cancel{hb}}{\cancel{h}} \\ &= f_x(x_0, y_0) \cdot \mathbf{a} + f_y(x_0, y_0) \cdot \mathbf{b} = \nabla f(x_0, y_0) \cdot \mathbf{u}. \end{aligned}$$

Letting $h \rightarrow 0$, we get $D_{\mathbf{u}}f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \mathbf{u}$.

Q09.8. Let $f(x, y, z) = xy^2z^3$ and $P = (1, -2, 1)$. Find the rate of change of f at P in the direction

$$\mathbf{u} = \left\langle \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle. \quad \|\mathbf{u}\| = 1.$$

[Remark: The question is the same as asking to find the directional derivative $D_{\mathbf{u}}f(P)$].

Solution: Given $f(x, y, z) = xy^2z^3$ and $P = (1, -2, 1)$.

Then $\nabla f = \langle f_x, f_y, f_z \rangle = \langle y^2z^3, 2xyz^3, 3xy^2z^2 \rangle$.

Hence, $\nabla f|_{(1,-2,1)} = \langle 4, -4, 12 \rangle$.

$$\nabla f(1, -2, 1)$$

Let $\mathbf{u} = \langle \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \rangle$. Note that $\|\mathbf{u}\| = 1$.

Thus, the rate of change of f at P in the direction $\mathbf{u}$ is:

$$D_{\mathbf{u}}f|_{(1,-2,1)} = \nabla f|_{(1,-2,1)} \cdot \mathbf{u}$$

$$\begin{aligned} D_{\mathbf{u}}f(1, -2, 1) &= \langle 4, -4, 12 \rangle \cdot \langle \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \rangle \\ &= 4 \cdot \frac{1}{\sqrt{3}} + (-4) \cdot (-\frac{1}{\sqrt{3}}) + 12 \cdot \frac{1}{\sqrt{3}} = \frac{20}{\sqrt{3}}. \end{aligned}$$

Q09.9. Let $\mathbf{a}$ be a positive number. Consider the function

$$f(x, y) = (x^2 + \mathbf{a}) e^{2x+3y^2}.$$

It is given that the directional derivative of $\mathbf{f}$ at the point $(-6, 2)$ in the direction of the vector $\mathbf{i} + 2\mathbf{j}$ is $216\sqrt{5}$. Find the value of $\mathbf{a}$.

Solution. Given $f(x, y) = (x^2 + a)e^{2x+3y^2}$,

the point $P = (-6, 2)$, and a vector $\mathbf{v} = \mathbf{i} + 2\mathbf{j}$. Then

$$f_x = 2x \cdot e^{2x+3y^2} + (x^2 + a) \cdot e^{2x+3y^2} \cdot 2,$$

$$f_y = (x^2 + a) \cdot 6y \cdot e^{2x+3y^2}.$$

Thus

$$\begin{aligned} f_x(-6, 2) &= 2(-6) \cdot e^{2 \cdot (-6) + 3 \cdot 2^2} + ((-6)^2 + a) \cdot e^{2 \cdot (-6) + 3 \cdot 2^2} \cdot 2 \\ &= -12 + (36 + a) \cdot 2 = \underline{60 + 2a}, \end{aligned}$$

$$\begin{aligned} f_y(-6, 2) &= (36 + a) \cdot 6 \cdot 2 \cdot e^{2 \cdot (-6) + 3 \cdot 2^2} \\ &= (36 + a) \cdot 12 = \underline{432 + 12a}. \end{aligned}$$

Hence $\nabla f(-6, 2) = (60 + 2a)\mathbf{i} + (432 + 12a)\mathbf{j}$.

The unit vector in the direction of $\mathbf{v} = \mathbf{i} + 2\mathbf{j}$ is $\|\mathbf{u}\| = 1$

$$\underline{\mathbf{u}} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{\mathbf{i} + 2\mathbf{j}}{\|\mathbf{i} + 2\mathbf{j}\|} = \frac{\mathbf{i} + 2\mathbf{j}}{\sqrt{1^2 + 2^2}} = \underline{\frac{1}{\sqrt{5}}\mathbf{i} + \frac{2}{\sqrt{5}}\mathbf{j}}.$$

Thus the directional derivative is:

$$\begin{aligned} D_{\mathbf{u}}f(-6, 2) &= \nabla f(-6, 2) \cdot \mathbf{u} \\ &= (60 + 2a) \cdot \frac{1}{\sqrt{5}} + (432 + 12a) \cdot \frac{2}{\sqrt{5}} \\ &= \underline{\frac{924 + 26a}{\sqrt{5}}}. \end{aligned}$$

Then

$$\begin{aligned} D_{\mathbf{u}}f(-6, 2) &= 216\sqrt{5} \implies \frac{924 + 26a}{\sqrt{5}} = 216\sqrt{5} \\ &\implies a = \frac{216 \cdot 5 - 924}{26} = \frac{156}{26} = \underline{\underline{6}}. \end{aligned}$$

Consider a differentiable function $f(x, y)$ and a point $P(x_o, y_o)$.

Question: At the given point $P(x_o, y_o)$, in which direction does f increase fastest?

Let $\mathbf{u} = a\mathbf{i} + b\mathbf{j}$ be a unit vector. (i.e., $\|\mathbf{u}\| = 1$.) Then

$$\begin{aligned}\underline{D_{\mathbf{u}}f(x_o, y_o)} &= \nabla f(x_o, y_o) \cdot \mathbf{u} \\ &= \|\nabla f(x_o, y_o)\| \|\mathbf{u}\| \cos \theta \\ &= \|\underline{\nabla f(x_o, y_o)}\| \cos \theta,\end{aligned}$$



where θ is the angle between $\nabla f(x_o, y_o)$ and $\mathbf{u}$.

Since the largest value of $\cos \theta$ is 1 which occurs when $\theta = 0$. Thus the largest possible value of $D_{\mathbf{u}}f(x_o, y_o)$ occurs when $\mathbf{u}$ is in the same direction as $\nabla f(x_o, y_o)$.

Since $\|\mathbf{u}\| = 1$, it follows that

$$\mathbf{u} = \frac{\nabla f(\mathbf{x}_o, \mathbf{y}_o)}{\|\nabla f(\mathbf{x}_o, \mathbf{y}_o)\|},$$

which gives the direction in which f increases most rapidly at the point $\mathbf{P}(\mathbf{x}_o, \mathbf{y}_o)$.

Remark. Similar argument shows that

$$\cos \theta = -1$$

$$\mathbf{u} = -\frac{\nabla f(\mathbf{x}_o, \mathbf{y}_o)}{\|\nabla f(\mathbf{x}_o, \mathbf{y}_o)\|}$$

gives the direction in which f decreases most rapidly at the point $\mathbf{P}(\mathbf{x}_o, \mathbf{y}_o)$.

Q09.10. Consider the function $f(x, y) = 2ye^{-x} + 1$ and the point $P(0, -2)$.

(a) Find the directional derivative of f at P in the direction of each of the following vectors:

(i) $\frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}$, (ii) $12\mathbf{i} + 5\mathbf{j}$.

(b) Find the direction that gives the *largest possible* directional derivative of f at P .

(c) Find the direction along which f decreases most rapidly at P .

(d) Find the directions along which f exhibits no change at P .

Solution: (a) Given that $f(x, y) = 2ye^{-x} + 1$. Thus,

$$f_x = 2ye^{-x} \cdot (-1) = -2ye^{-x}, \quad f_y = 2e^{-x}.$$

Then at $P(0, -2)$, we have

$$f_x(0, -2) = -2(-2)e^{-0} = 4 \quad \text{and} \quad f_y(0, -2) = 2e^{-0} = 2.$$

$$\text{Thus } \nabla f(0, -2) = f_x(0, -2)\mathbf{i} + f_y(0, -2)\mathbf{j} = \underline{4\mathbf{i} + 2\mathbf{j}}.$$

(i) $\mathbf{u} = \frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}$ is a unit vector. Thus,

$$\begin{aligned} D_{\mathbf{u}}f(0, -2) &= \nabla f(0, -2) \cdot \mathbf{u} \\ &= (4\mathbf{i} + 2\mathbf{j}) \cdot \left(\frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}\right) \\ &= 4 \times \frac{1}{\sqrt{2}} + 2 \times \frac{1}{\sqrt{2}} = \underline{3\sqrt{2}}. \end{aligned}$$

(ii) The unit vector in the direction of $12\mathbf{i} + 5\mathbf{j}$ is

$$\mathbf{u} = \frac{12\mathbf{i} + 5\mathbf{j}}{\sqrt{12^2 + 5^2}} = \frac{12}{13}\mathbf{i} + \frac{5}{13}\mathbf{j}.$$

Thus,

$$\begin{aligned} D_{\mathbf{u}}f(0, -2) &= \nabla f(0, -2) \cdot \mathbf{u} \\ &= (4\mathbf{i} + 2\mathbf{j}) \cdot \left(\frac{12}{13}\mathbf{i} + \frac{5}{13}\mathbf{j}\right) \\ &= 4 \times \frac{12}{13} + 2 \times \frac{5}{13} = \frac{58}{13}. \end{aligned}$$

(b) Find the direction that gives the *largest possible* directional derivative of f at P .

Solution: Let $\mathbf{u} = a\mathbf{i} + b\mathbf{j}$ be a unit vector. (i.e., $\|\mathbf{u}\| = 1$.) Then

$$\begin{aligned} D_{\mathbf{u}}f(0, -2) &= \nabla f(0, -2) \cdot \mathbf{u} \\ &= \|\nabla f(0, -2)\| \|\mathbf{u}\| \cos \theta \end{aligned}$$

where θ is the angle between $\nabla f(0, -2)$ and $\mathbf{u}$.

Since the largest value of $\cos \theta$ is 1 which occurs when $\theta = 0$, this means that the largest possible value of $D_{\mathbf{u}}f(0, -2)$ occurs when $\mathbf{u}$ is in the same direction as $\nabla f(0, -2)$. Since $\|\mathbf{u}\| = 1$, it follows that

$$\mathbf{u} = \frac{\nabla f(0, -2)}{\|\nabla f(0, -2)\|} = \frac{4\mathbf{i} + 2\mathbf{j}}{\|4\mathbf{i} + 2\mathbf{j}\|} = \frac{4\mathbf{i} + 2\mathbf{j}}{\sqrt{4^2 + 2^2}} = \frac{2}{\sqrt{5}}\mathbf{i} + \frac{1}{\sqrt{5}}\mathbf{j}.$$

(c) Find the direction along which f decreases most rapidly at P .

Solution: Recall from part (b):

$$\begin{aligned} D_{\mathbf{u}}f(0, -2) &= \nabla f(0, -2) \cdot \mathbf{u} \\ &= \|\nabla f(0, -2)\| \|\mathbf{u}\| \cos \theta \end{aligned}$$

where θ is the angle between $\nabla f(0, -2)$ and $\mathbf{u}$.

$D_{\mathbf{u}}f(0, -2)$ is most negative when $\cos \theta = -1$, i.e., when $\theta = \pi$. This happens when $\mathbf{u}$ is in opposite direction to $\nabla f(0, -2)$. Thus

$$\mathbf{u} = \frac{-\nabla f(0, -2)}{\|-\nabla f(0, -2)\|} = \frac{-(4\mathbf{i} + 2\mathbf{j})}{\|-(4\mathbf{i} + 2\mathbf{j})\|} = -\frac{2}{\sqrt{5}}\mathbf{i} - \frac{1}{\sqrt{5}}\mathbf{j}.$$

(d) Find the directions along which $\mathbf{f}$ exhibits no change at $\mathbf{P}(0, -2)$.

Solution: Let $\mathbf{u} = a\mathbf{i} + b\mathbf{j}$ be a unit vector, along which $\mathbf{f}$ exhibits no change at $\mathbf{P}$. Recall from part (b):

$$\begin{aligned} D_{\mathbf{u}}f(0, -2) &= \nabla f(0, -2) \cdot \mathbf{u} \\ &= \|\nabla f(0, -2)\| \|\mathbf{u}\| \cos \theta \end{aligned}$$

where θ is the angle between $\nabla f(0, -2)$ and $\mathbf{u}$. Thus

$$\begin{aligned} D_{\mathbf{u}}f(0, -2) &= 0 \\ \iff \nabla f(0, -2) \cdot \mathbf{u} &= 0 \\ \iff (4\mathbf{i} + 2\mathbf{j}) \cdot (a\mathbf{i} + b\mathbf{j}) &= 0 \\ \iff 4a + 2b = 0 \iff \underline{b = -2a}. &\quad (1) \end{aligned}$$

Now

$$\| \mathbf{u} \| = 1 \implies \| \mathbf{a}\mathbf{i} + \mathbf{b}\mathbf{j} \|^2 = 1 \implies \underline{a^2 + b^2 = 1}.$$

Together with (1) that $\mathbf{b} = -2\mathbf{a}$, we have

$$a^2 + (-2a)^2 = 1 \implies 5a^2 = 1 \implies \underline{a = \pm \frac{1}{\sqrt{5}}}.$$

When $a = \frac{1}{\sqrt{5}}$, $b = -2a = -\frac{2}{\sqrt{5}}$, $\therefore \mathbf{u} = \frac{1}{\sqrt{5}}\mathbf{i} - \frac{2}{\sqrt{5}}\mathbf{j}$. When
 $a = -\frac{1}{\sqrt{5}}$, $b = -2a = \frac{2}{\sqrt{5}}$, $\therefore \mathbf{u} = -\frac{1}{\sqrt{5}}\mathbf{i} + \frac{2}{\sqrt{5}}\mathbf{j}$.

Thus the directions are along the unit vectors $\frac{1}{\sqrt{5}}\mathbf{i} - \frac{2}{\sqrt{5}}\mathbf{j}$ and
 $-\frac{1}{\sqrt{5}}\mathbf{i} + \frac{2}{\sqrt{5}}\mathbf{j}$.

Recall the Chain Rule: Take $f(x, y, z)$, where

$$x = x(t), \quad y = y(t), \quad z = z(t). \quad (1)$$

Regarding f as a function of t , we have

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial f}{\partial z} \cdot \frac{dz}{dt}. \quad (2)$$

Take the vector function $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$. Then

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle, \quad \mathbf{r}'(t) = \left\langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right\rangle.$$

Then (2) can be written as:

$$\frac{df}{dt} = \nabla f \cdot \mathbf{r}'(t),$$

with ∇f evaluated at the point with position vector $\mathbf{r}(t)$.

Remark. Similar formula holds for $f(x, y)$.

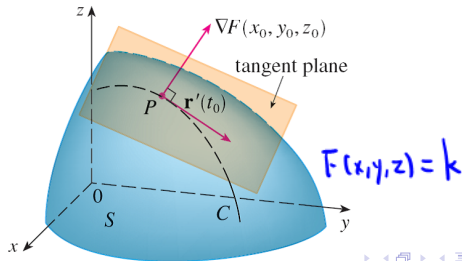
Remark. In Q09.03, the rate of change of temperature $\frac{dT}{dt}$ experienced by the insect is thus also given by the dot product $\nabla T \cdot \mathbf{r}'(t)$, where the $\mathbf{r}'(t)$ is the velocity vector of the insect at time t .

Consider a function $F(x, y, z)$, and for a constant k , consider the level surface S given by

$$\underline{F(x, y, z) = k.}$$

Take a point $P = (x_0, y_0, z_0)$ on S (so that $F(x_0, y_0, z_0) = k$). Consider all the curves C lying on S and passing through P . The tangent vectors to such curves C at P (and with tail at P) form a plane passing through P , known as the tangent plane Π to S at P .

$$C: \mathbf{r}(t)$$



• **Fact:** $\nabla F(x_0, y_0, z_0) \perp \Pi$ (i.e. $\nabla F(x_0, y_0, z_0)$ is a normal vector to Π .)

To see this, we take an arbitrary curve C lying on S , passing through P and defined by the vector function $\mathbf{r}(t) = \langle \mathbf{x}(t), \mathbf{y}(t), \mathbf{z}(t) \rangle$ with $\mathbf{r}(t_0) = \mathbf{P}$. Note that $\mathbf{r}'(t_0)$ is the tangent vector to C at P . Since C lies on S , we have

$$\underline{F(\mathbf{x}(t), \mathbf{y}(t), \mathbf{z}(t)) = k} \quad \text{for all } t.$$

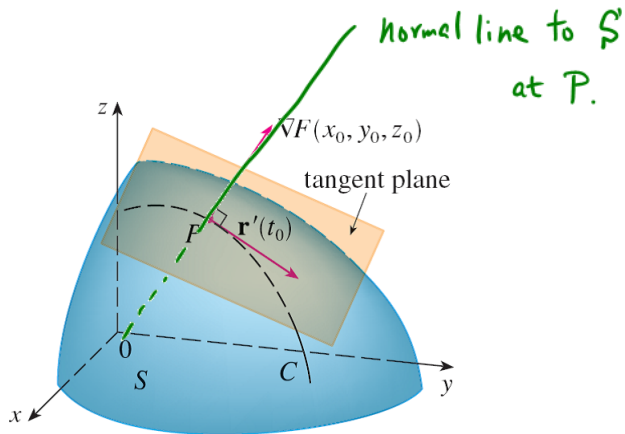
F
 $x \ y \ z$
 t

Differentiating both sides with respect to t and using the Chain Rule on the left hand side, we get $\nabla F \cdot \mathbf{r}'(t) = 0$.

At $t = t_0$, we have $\frac{dF}{dt} =$

$$\nabla F(x_0, y_0, z_0) \cdot \mathbf{r}'(t_0) = 0 \implies \nabla F(x_0, y_0, z_0) \perp \mathbf{r}'(t_0).$$

Since Π consists of tangent vectors of the form $\mathbf{r}'(t_0)$, it follows that $\nabla F(x_0, y_0, z_0) \perp \Pi$.



$$F(x_0, y_0, z_0) = k, \nabla F(x_0, y_0, z_0) \cdot \mathbf{r}'(t_0) = 0.$$

Consequently, the **tangent plane to the level surface** $F(x, y, z) = k$ at (x_o, y_o, z_o) is given by the equation

$$\underline{\nabla F(x_o, y_o, z_o)} \cdot \underline{\langle x - x_o, y - y_o, z - z_o \rangle} = 0$$

or equivalently,

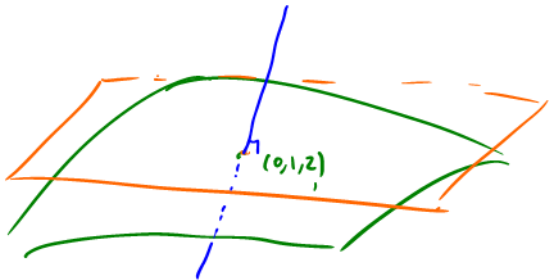
$$F_x(x_o, y_o, z_o)(x - x_o) + F_y(x_o, y_o, z_o)(y - y_o) + F_z(x_o, y_o, z_o)(z - z_o) = 0.$$

Remark. The **normal line** to $\mathbf{S}$ at $\mathbf{P}$ is simply the line passing through $\mathbf{P}$ and perpendicular to (the tangent plane of) $\mathbf{S}$ at $\mathbf{P}$ (and thus parallel to $\underline{\nabla F(\mathbf{P})}$).

Q10.1. Consider the surface in the **xyz**-space defined by the equation

$$\cos(\pi x) - x^2 y + e^{xz} + yz = 4.$$

Find equations for the tangent plane and normal line to the surface at the point (0, 1, 2).



Solution: S is the level surface $f(x, y, z) = 4$,

where $f(x, y, z) = \cos(\pi x) - x^2 y + e^{xz} + yz$.

Given the point $P(0, 1, 2)$ on S .

$$\begin{aligned}\nabla f &= \langle f_x, f_y, f_z \rangle \\ &= \langle -\pi \sin(\pi x) - 2xy + ze^{xz}, -x^2 + 2, xe^{xz} + y \rangle\end{aligned}$$

$$\nabla f(0, 1, 2) = \langle 2, 2, 1 \rangle$$

$\therefore$ the tangent plane is

$$\nabla f(0, 1, 2) \cdot \langle x-0, y-1, z-2 \rangle = 0$$

$$\langle 2, 2, 1 \rangle \cdot \langle x, y-1, z-2 \rangle = 0$$

$$2x + 2(y-1) + (z-2) = 0$$

$$2x + 2y + z = 4$$

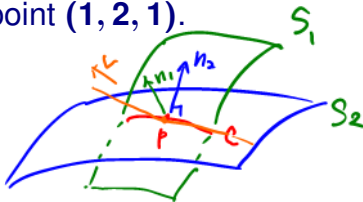
Normal line: $x = 0 + 2t$, $y = 1 + 2t$, $z = 2 + t$.

$$y = 1 + 2t$$

Q10.2. Let $\mathbf{C}$ be the curve of the intersection of the two surfaces $\mathbf{x}^3 + 2\mathbf{xy} + \mathbf{yz} = 7$ and $3\mathbf{x}^2 - \mathbf{yz} = 1$.

Find parametric equations of the tangent line to $\mathbf{C}$ at $(1, 2, 1)$.

[Suggestion: First, for each of the two surfaces, find a normal vector of the surface at the point $(1, 2, 1)$.



Solution: The surface $\mathbf{S}_1$ with equation $x^3 + 2xy + yz = 7$ is the level surface

$$f(x, y, z) = 7, \text{ where } f(x, y, z) = x^3 + 2xy + yz.$$

$$\text{Now } \nabla f = \langle f_x, f_y, f_z \rangle = \langle 3x^2 + 2y, 2x + z, y \rangle.$$

Then a normal vector to $\mathbf{S}_1$ at the point $(1, 2, 1)$ is:

$$\underline{\mathbf{n}_1} = \nabla f(1, 2, 1) = \langle 3 \cdot 1^2 + 2 \cdot 2, 2 \cdot 1 + 1, 2 \rangle = \underline{\langle 7, 3, 2 \rangle}.$$

Similarly, the surface $\mathbf{S}_2 : 3x^2 - yz = 1$ is the level surface

$$g(x, y, z) = 1, \text{ where } g(x, y, z) = 3x^2 - yz.$$

$$\text{Now } \nabla g = \langle g_x, g_y, g_z \rangle = \langle 6x, -z, -y \rangle.$$

Then a normal vector to $\mathbf{S}_2$ at the point $(1, 2, 1)$ is:

$$\mathbf{n}_2 = \nabla g(1, 2, 1) = \langle 6, -1, -2 \rangle.$$

Note that the curve of intersection $\mathbf{C}$ of the two surfaces $\mathbf{S}_1$ and $\mathbf{S}_2$ lies on both $\mathbf{S}_1$ and $\mathbf{S}_2$. Thus any tangent vector to $\mathbf{C}$ at $(1, 2, 1)$ lies on both tangent planes of $\mathbf{S}_1$ and $\mathbf{S}_2$ at $(1, 2, 1)$, and thus it is perpendicular to both $\mathbf{n}_1$ and $\mathbf{n}_2$. In particular, a tangent vector to the curve $\mathbf{C}$ at $(1, 2, 1)$ is given by

$$\mathbf{v} = \mathbf{n}_1 \times \mathbf{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 7 & 3 & 2 \\ 6 & -1 & -2 \end{vmatrix} = \langle -4, 26, -25 \rangle.$$

Therefore parametric equations of the tangent line to $\mathbf{C}$ at $(1, 2, 1)$ is:

$$x = 1 - 4t, \quad y = 2 + 26t, \quad z = 1 - 25t.$$

Remark. For a function $g(x, y)$, its graph is the surface S defined by the equation $z = g(x, y)$. Note that

$$z = g(x, y) \iff g(x, y) - z = 0.$$

Thus S can also be regarded as the level surface

$$f(x, y, z) = 0, \quad \text{where } f(x, y, z) = g(x, y) - z.$$

Q10.2.5. (Supplementary Exercise) Recall that the equation of the tangent plane to the graph $z = g(x, y)$ at (x_0, y_0) is given by

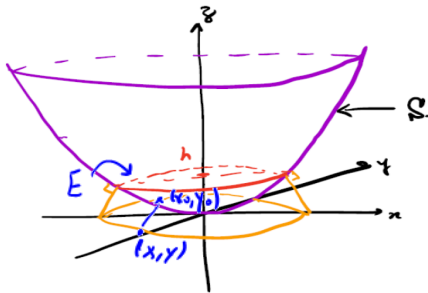
$$z = g(x_0, y_0) + g_x(x_0, y_0)(x - x_0) + g_y(x_0, y_0)(y - y_0). \quad (*)$$

Derive (*) by regarding the graph of $g(x, y)$ as a level surface.

Q10.3. Let $\mathbf{S}$ be the surface given by the equation $z = \mathbf{a}x^2 + \mathbf{b}y^2$, where $\mathbf{a}$ and $\mathbf{b}$ are some positive constants.

Suppose normal lines are drawn at all the points on $\mathbf{S}$ that are at height $\mathbf{h}$ above the $\mathbf{xy}$ -plane.

Find an equation of the curve $\mathbf{C}$ (in terms of $\mathbf{x}$ and $\mathbf{y}$) consisting of all the intersection points of these normal lines and the $\mathbf{xy}$ -plane.



Solution: Note that

$$z = ax^2 + by^2 \iff ax^2 + by^2 - z = 0.$$

Thus **S** is the level surface

$$f(x, y, z) = 0, \quad \text{where } f(x, y, z) = ax^2 + by^2 - z.$$

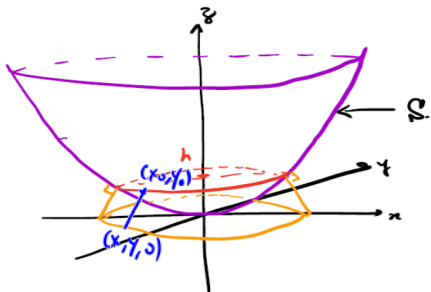
Thus a normal vector to **S** at $(x, y, z) \in \mathbf{S}$ is given by

$$\nabla f(x, y, z) = \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix} = \begin{pmatrix} 2ax \\ 2by \\ -1 \end{pmatrix}.$$

A point on **S** at height **h** above the **xy**-plane is of the form (x_o, y_o, h) with $ax_o^2 + by_o^2 = h$. Such points (x_o, y_o, h) form a curve, which we denote by **E**.

Then a normal vector to $\mathbf{S}$ at such $(\mathbf{x}_o, \mathbf{y}_o, \mathbf{h})$ is given by

$$\nabla f(\mathbf{x}_o, \mathbf{y}_o, \mathbf{h}) = \begin{pmatrix} 2a\mathbf{x}_o \\ 2b\mathbf{y}_o \\ -1 \end{pmatrix},$$



and thus the normal line to $\mathbf{S}$ at $(\mathbf{x}_o, \mathbf{y}_o, \mathbf{h})$ is given by

$$\mathbf{x} = \mathbf{x}_o + 2a\mathbf{x}_o t, \quad \mathbf{y} = \mathbf{y}_o + 2b\mathbf{y}_o t, \quad \mathbf{z} = \mathbf{h} - t.$$

The normal line intersects the $\mathbf{xy}$ -plane at a point where $\mathbf{z} = \mathbf{h} - t = 0$, i.e. $t = \mathbf{h}$, and the point is

$$(\mathbf{x}, \mathbf{y}, 0) = (\mathbf{x}_o(1 + 2a\mathbf{h}), \mathbf{y}_o(1 + 2b\mathbf{h}), 0).$$

The curve **C** is the locus of the points

$$(x, y, 0) = (x_o(1 + 2ah), y_o(1 + 2bh), 0)$$

when the point (x_o, y_o, h) varies along the curve **E**.

Since points (x_o, y_o, h) on **E** satisfy the equation

$$ax_o^2 + by_o^2 = h,$$

the corresponding points $(x, y, 0)$ on **C** satisfy the equation

$$\frac{ax^2}{(1 + 2ah)^2} + \frac{by^2}{(1 + 2bh)^2} = h.$$

